

2024



# Creating Immersive VR Games Using Amazon GameLift with Virtuix

Learn how Virtuix in the virtual reality industry built an immersive gaming experience using Amazon GameLift.

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## Overview

You may have experienced virtual reality (VR) before—now imagine truly walking around in a digital world. [Virtuix Inc.](#) (Virtuix) makes a more immersive, at-home VR experience possible with its innovative omnidirectional treadmill, the Omni.

Using a variety of Amazon Web Services (AWS) solutions, Virtuix built a scalable backend infrastructure and moved its product from beta testing to full production. “This will be the first complete, full-body VR entertainment system for the home—not just a headset, but a headset and treadmill that you can use to run around inside video games from your living room,” says Jan Goetgeluk, founder and CEO of Virtuix. “And it’s entirely powered by AWS.” On AWS, Virtuix is prepared to scale to meet the demand of players worldwide.



## Opportunity | Using AWS to Build Innovative VR Products for Virtuix

Founded in 2013, Virtuix is creating innovative products in the VR industry. After raising over 1 million dollars on Kickstarter, Virtuix created the Omni, which players can use to walk, run, and turn 360 degrees in VR. Virtuix’s first two products, the Omni Pro and the multiplayer Omni Arena, are popular attractions at entertainment venues and esports events. In fact, more than 4,000 Omni Pro units have been shipped around the world.

Virtuix’s goal, however, was always to create a consumer product designed for personal use. After iterating, and ultimate success of the commercial versions, Virtuix finally unveiled plans for the



intended for at-home use—in October 2020. By 2024, after three phases of product verification testing, beta units of the Omni One were completed and shipped out.

Virtuix has been all-in on AWS from the beginning, using AWS to build the software for its products and develop its own VR games. The Virtuix team chose AWS primarily for its scalability; it needed a cloud service provider that could grow alongside the company. Plus, developing the Omni was a big undertaking that required services for hardware, software, websites, and games—having a cloud service provider that could seamlessly connect all these moving parts was crucial for Virtuix. “AWS is built into the fabric of our engineering process,” says Goetgeluk. “The support is the highest quality; the AWS team helps us find the right services and answers any questions we have.”

## Solution | Achieving 99% Uptime During Omni One Beta Testing

Virtuix doesn’t only manufacture innovative VR hardware; it has also created multiple VR games optimized for the Omni. To make sure that players can connect to multiplayer games all around the world, Virtuix uses [Amazon GameLift](#), which deploys and manages dedicated game servers hosted in the cloud, on premises, or through hybrid deployments. Using Amazon GameLift, Virtuix gained a low-latency solution with the flexibility to scale up and down as needed. “The Omni really comes to life when you use it with other people,” says Cameron Slayter, creative director at Virtuix. “As the Omnis are shipped to different locations, we can spin up a server anywhere to keep the latency low for all our players.”

Virtuix uses multiple AWS services to keep its backend development running smoothly. Primarily, its infrastructure is built on [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. It also uses [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed service that makes it simple for developers to run Kubernetes on AWS without needing to install and operate their own Kubernetes clusters. These services are not only scalable, which helps Virtuix meet rising demand for its products, but they are also reliable. In fact, Virtuix achieved 99 percent uptime with zero crashes during beta testing of the Omni One. “As we start to scale up, we’re going to rely on Amazon EC2 and Amazon EKS to meet demand,” says Jerry Bogedain, director of software at Virtuix.

On AWS, the Virtuix team doesn’t have to worry about the day-to-day maintenance of managing infrastructure. In fact, DevOps only accounts for 20–30 hours of work per month at Virtuix. Rather, the team can focus on optimizing its products and creating new innovations in VR. “The ease of use and quick setup time of AWS solutions means that we can try new things and pivot when we need to,” says Bogedain. “That really is the definition of a solution.”

Virtuix uses AWS products to improve the customer experience as well. For example, using [Amazon SES](#) (Amazon SES), a reliable and scalable service that helps companies communicate with customers, Virtuix



can reach its customers at multiple touchpoints. Amazon SES makes it simple for customers to change their passwords, buy games on Virtuix's online store, and receive follow-up emails and shopping notifications.

## Outcome | Scaling Up to Meet Enthusiastic Demand

Using AWS, Virtuix brought its innovative product from idea to reality. With beta units of the Omni One already shipped out, the company plans to release the solution to the general market in Q2 2024, ramping up production while maintaining its focus on quality and software development. Virtuix is targeting 50 percent gross margin after launch.

As the Omni One user base grows, the need for more infrastructure increases as well. Virtuix will expand its use of AWS products as the company plans to scale from less than 100 customers to more than 10,000 by the end of 2024. In particular, it will incorporate data analytics using [Amazon Redshift](#)—which uses SQL to analyze structured and semistructured data across data warehouses, operational databases, and data lakes—to build business intelligence and better understand its players. In February 2024, the team will also launch more than 30 third-party games for the Omni One on its own game store.

“People get quite excited when they see what they can do with our product,” says Goetgeluk. “The demand is there, and now the product is there. On AWS, we’re ready to scale.”

## About Virtuix

Virtuix is a virtual reality gaming company that manufactures omnidirectional treadmills for immersive, 360-degree experiences that are popular in entertainment centers worldwide. In 2024, it started shipping Omni One, its first system designed for the home.

## AWS Services Used

### Amazon GameLift

Amazon GameLift deploys and manages dedicated game servers hosted in the cloud, on-premises, or through hybrid deployments. Amazon GameLift provides a low-latency and low-cost solution that scales with fluctuating player demand.

[Learn more »](#)

### Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute portfolio with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase



model to help you best match the needs of your workload.

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## Amazon EKS

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed service that makes it easy for you to run Kubernetes on AWS without needing to install and operate your own Kubernetes clusters.

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## Amazon SES

Amazon Simple Email Service (Amazon SES) lets you reach customers confidently without an on-premises Simple Mail Transfer Protocol (SMTP) email server using the Amazon SES API or SMTP interface.

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